Medical Infusion Pump FMEA Case Study

Reliability Engineering Portfolio

Project Overview

This case study applies Failure Mode and Effects Analysis (FMEA) to a medical infusion pump. The analysis identifies potential failure modes, considers their effects and causes, assigns Severity, Occurrence, and Detection ratings, and uses the resulting RPN to prioritize recommended actions.

Objective

Identify failure modes that could affect pump operation, medication delivery, alarm availability, or physical integrity, and recommend actions to reduce or control associated risks.

FMEA Results

Item	Failure Mode	Effect	Cause	S	O	D	RPN	Recommended Action
Battery	No longer holds charge	Reduced battery capacity and operational readiness	Normal wear	8	9	2	144	Add an alarm/indicator that notifies the user when fully charged and at 25% battery level. Intended to discourage prolonged charging and help maintain operational readiness.
Touchscreen	Becomes unresponsive	User input may be missed, locked, or inaccurately registered	Human error	7	9	3	189	Encourage staff to wipe the touchscreen with a dry, tight-weave microfiber cloth after disinfecting and before entering values.
Flow-rate sensor	Drifts out of calibration	Medication dosage becomes inaccurate	Mechanical wear	9	8	8	576	Schedule routine preventive-maintenance checks; analyze data to determine the extent of sensor drift before adjustment; verify accuracy every six months.
Power supply	Fails; unit shuts down	Unit suddenly shuts down	Battery degradation / backup failure	9	1	8	72	Add monitoring/alerts for battery-backup faults and battery health.
Alarm speaker	Fails	Pump continues operating but alarm cannot be heard	Hardware failure, software error, or configuration setting	8	8	7	448	Check speaker operation and volume, then run software diagnostics. If a hardware/electrical fault is identified, remove the pump from service and replace it with a functional unit while the fault is corrected.
Housing	Cracks	Potential cosmetic or structural damage; physical integrity may be affected	Accidental impact, tubing misalignment/stress, or chemical degradation from harsh cleaning agents	8	9	7	441	Remove the pump from service and replace the damaged housing/pump as appropriate.

Risk Prioritization

• Flow-rate sensor — RPN 576

• Alarm speaker — RPN 448

• Housing — RPN 441

• Touchscreen — RPN 189

• Battery — RPN 144

• Power supply — RPN 72

The highest RPN is 576 for flow-rate sensor calibration drift, followed by alarm-speaker failure (448) and housing cracks (441). These values are retained from the original exercise and should not be treated as a validated clinical risk ranking without organization-specific criteria and supporting data.

Limitations.

This is an educational reliability-engineering case study, not a validated clinical risk assessment. The ratings reflect the assumptions and information available for the exercise.

References and Evidence

The original analysis referenced FDA information and other research for selected failure modes, but specific source details were not included in the supplied document. Those citations should be added to references.md before publication.